

CODSOFT INTERNSHIP

Python Mini Projects

Internship Project Presentation

Calculator • Password Generator • Rock-Paper-Scissors

PRESENTED BY

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ACADEMIC DETAILS

B.Tech CSE

Semester III

PYTHON

**MINI
PROJECTS**

01 INPUT

02 LOGIC

03 OUTPUT

04 RANDOM

Organization: Codsoft

Internship Project Overview

Three Python applications developed to practice core programming concepts.

02

CALCULATOR

Basic arithmetic using two numbers and an operation choice.

03

PASSWORD GENERATOR

Random password creation based on user-defined length.

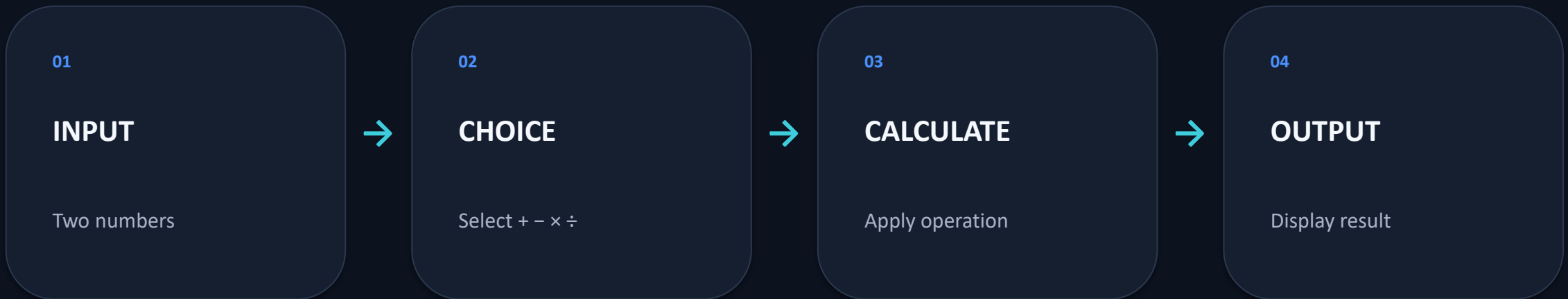
04

ROCK-PAPER-SCISSORS

Interactive game using random computer selection and game logic.

Simple Calculator

A command-line calculator for basic arithmetic operations.



Learning focus: user input • type conversion • operators • if/elif/else • basic error handling

Password Generator

Generate a random password from a user-specified length.

USER INPUT

Enter desired length

Example: 12 characters



Length controls output size.

GENERATION

**Random
characters**

Letters + numbers + symbols

random module

OUTPUT

**Strong-looking
random password**

Displayed directly on the screen.

Rock-Paper-Scissors

A simple user-versus-computer game with repeat-play support.

ROCK

Beats **SCISSORS**

PAPER

Beats **ROCK**

SCISSORS

Beats **PAPER**

User choice → Computer random choice → Compare rules → Win / Lose / Tie

Optional extension: score tracking and “Play Again” loop.

Python Concepts Used

The projects combine fundamental programming concepts into complete mini applications.

Python

Core language used for all three tasks.

Input / Output

Collect user choices and display results.

Conditions

Control arithmetic and game outcomes.

Random Module

Create random passwords and computer choices.

Loops

Support repeated gameplay and interaction.

Functions

Organize logic into reusable blocks.

Skills Developed

Practical outcomes from building and testing the projects.

PROBLEM SOLVING

Breaking requirements into small logical steps.

PROGRAMMING LOGIC

Using conditions, loops and operators effectively.

USER INTERACTION

Designing clear input and output flows.

RANDOMIZATION

Applying Python's random functionality.

DEBUGGING

Finding and fixing input and logic errors.

PROJECT CONFIDENCE

Turning simple ideas into working applications.

Conclusion & Future Scope

A foundation for moving from small scripts to larger Python applications.

KEY TAKEAWAY

**These projects strengthened
Python fundamentals and logical thinking.**

Future scope: graphical interfaces, stronger password rules, scoreboards, file handling and larger Python applications.

PROJECTS COMPLETED

- ✓ Calculator
- ✓ Password Generator
- ✓ Rock-Paper-Scissors

Thank you